

Chemistry I

Final Exam

10th November, 2022

6th Period

ID

MAJOR:

NAME:

Score distribution for each question

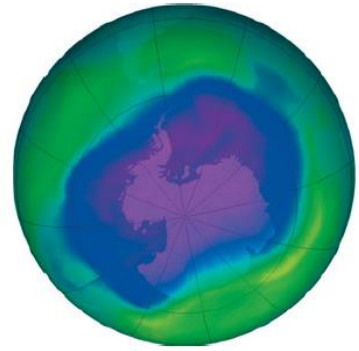
1.	3 %
2.	6 %
3.	3 %
4.	8 %
5.	8 %
6.	8 %
7.	4 %
8.	4 %
9.	3 %
10.	4 %
11.	4 %
12.	5 %

Total 60%

Grading

EXAM (60 %) + Assignment (30 %) + Class Attitude (10 %) = 100 %

1. A NASA satellite showed that in 2009 the ozone hole over Antarctica had a maximum surface area of 24.1 million km². The largest ozone hole on record occurred in 2006 and had a surface area of 29.6 million km². Calculate the difference in diameter (in meters) between the ozone hole in 2009 and in 2006.



***NOTE: Describe the solving Process**

▲ A layer of ozone gas (a form of oxygen) in the upper atmosphere protects Earth from harmful ultraviolet radiation in sunlight. Human-made chemicals react with the ozone and deplete it, especially over the Antarctic at certain times of the year (the so-called ozone hole). The region of low ozone concentration in 2006 (represented here by the dark purple color) was the largest on record.

2. In one transition, an electron in a hydrogen atom falls from the $n=3$ level to the $n=2$ level. In a second transition, an electron in a hydrogen atom falls from the $n=2$ level to the $n=1$ level. Compared to the radiation emitted by the first of these transitions, the radiation emitted by the second will have:

- (a) a lower frequency (b) a smaller energy per photon
- (c) a shorter wavelength (d) a longer wavelength

Describe why you choose the answer: ***NOTE: explain in detail as possible**

3. Promethium ($Z = 61$) is essentially nonexistent in nature; all of its isotopes are radioactive. Write balanced nuclear equations for the decomposition of Pm-147 by beta emission.

***NOTE: explain in detail as possible**

4. Calculate both the **radius (r_n)** and **energy (E_n)** using equation 11 and 12 below. eV (electron volt) is used as the unit that expresses the energy of electrons in atom. 1 [eV] means the amount of energy can be obtained when a particle having a charge, e , is accelerated by 1 [V]. 1 [MeV] is often used rather than 1 [eV]. 1 [MeV] equals to 10^6 [eV]. Length should be expressed using [$\text{\AA}$].

$$\hbar c = 197.323 \text{ [eV} \cdot \text{nm]} \approx 200 \text{ [eV} \cdot \text{nm]}, (1[\text{\AA}] = 0.1[\text{nm}] = 10^{-10}[\text{m}])$$

equation of fine structure constant (α): $\frac{e^2}{\hbar c} = \frac{1}{137}$

Einstein's equation of $E=mc^2 = 0.511 \times 10^6 \text{ [eV]}$

Equation 11: $r_n = \frac{\hbar^2}{me^2} n^2$, Equation 12: $E_n = -\frac{e^2}{2r} = -\frac{e^2 me^2}{2n^2 \hbar^2} = -\frac{me^4}{2\hbar^2} \times \frac{1}{n^2}$

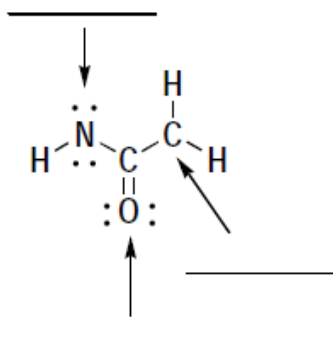
- 5.** Calculate Rydberg constant (R) using equations used in Question 5. *NOTE: explain in detail as possible

$$\hbar c = 197 \text{ [eV} \cdot \text{nm]} = 197 \times 10^{-9} \text{ [eV} \cdot \text{nm]}$$

$$R = \frac{me^4}{4\pi\hbar^3} =$$

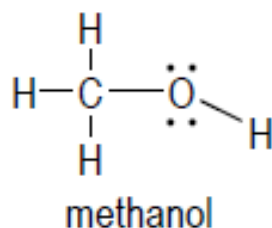
- 6.** Based on r_n and E_n obtained from Question 5, calculate energy and the radius when $n = 1, 2$, and 3 . Length should be expressed using [$\text{\AA}$]

- 7.** Calculate the formal charge for the indicated atoms in the following structure:

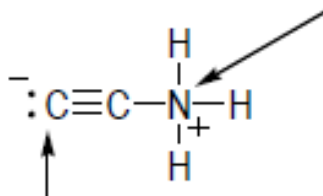


8. Draw three resonance structures for CS_3^{2-}

9. Using the following figure, show the bond dipole of the C-O bond and the O-H bond in methanol by using the dipole vector.



10. Predict the hybridizations, geometries, and bond angles for each of the atoms where indicated in the shown molecule.



11. Draw the orbital diagram for methane, CH_4 .

12. Draw the full Lewis formula and orbital diagram of formaldehyde, H_2CO .